When the Ball Holds the Pair, δS Is the Pair Mixing Entropy.

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Abstract

Paper 72 tracked pair δS on $R = 3$ balls that only touch the source sites. This note uses $R = 5$. Exact open-hop basis. Peschel is `mpmath eigsy` at 80 decimals. No hop LAPACK.

$$S_{\text{global}} = 2h(\alpha_A) + 2h(\alpha_B).$$

Midpoint enclosing ball: $f_S = 0.9907$, $f_E = 0.9881$. A one-site axial offset still contains both sites but leaks the heavier packet: $f_S = 0.886 = f_E$ to half a percent, not 1. A ball on A with B outside: $f_S = 0.382$, $f_E = 0.386$. Auditor $N = 11$: midpoint 0.994; a perpendicular offset stays at 0.990; miss 0.306. Primary enclosure confirmed. Axial $f = 1$ fails as leak.

Pair δS is pair mixing entropy times the enclosed mass fraction. Not Einstein.

Admission. *I pre-registered $f = 1$ on any offset that contains both sites. The axial offset refused. I did not move C_off.*